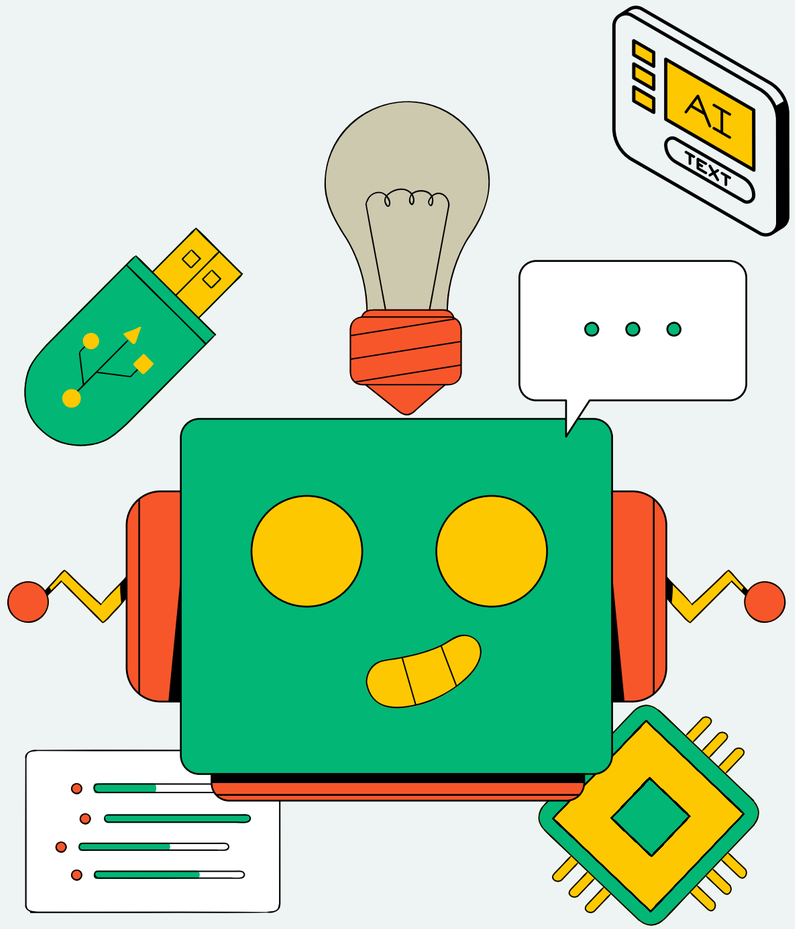

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

AI-BASED DRIVER DROWSINESS DETECTION SYSTEM

"BECAUSE ONE MOMENT OF SLEEP CAN CHANGE EVERYTHING"

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INTRODUCTION

Driver drowsiness is one of the major causes of road accidents worldwide. Fatigue reduces reaction time, awareness, and decision-making ability. Many drivers are unaware of their drowsiness until it is too late.

“The most dangerous thing on the road is not speed—it’s sleep.”

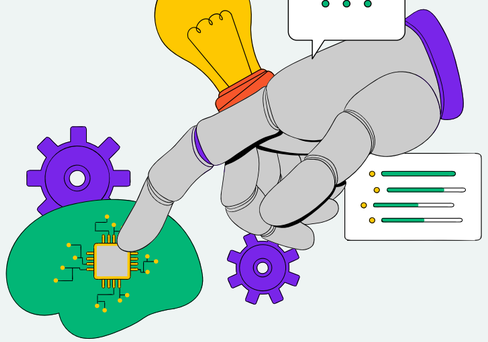
PROBLEM STATEMENT

Driver fatigue leads to a large number of accidents, especially during long-distance and night driving. Existing systems are expensive and not accessible to everyone.

“The problem is not that drivers fall asleep… it’s that they don’t realize when it starts.”

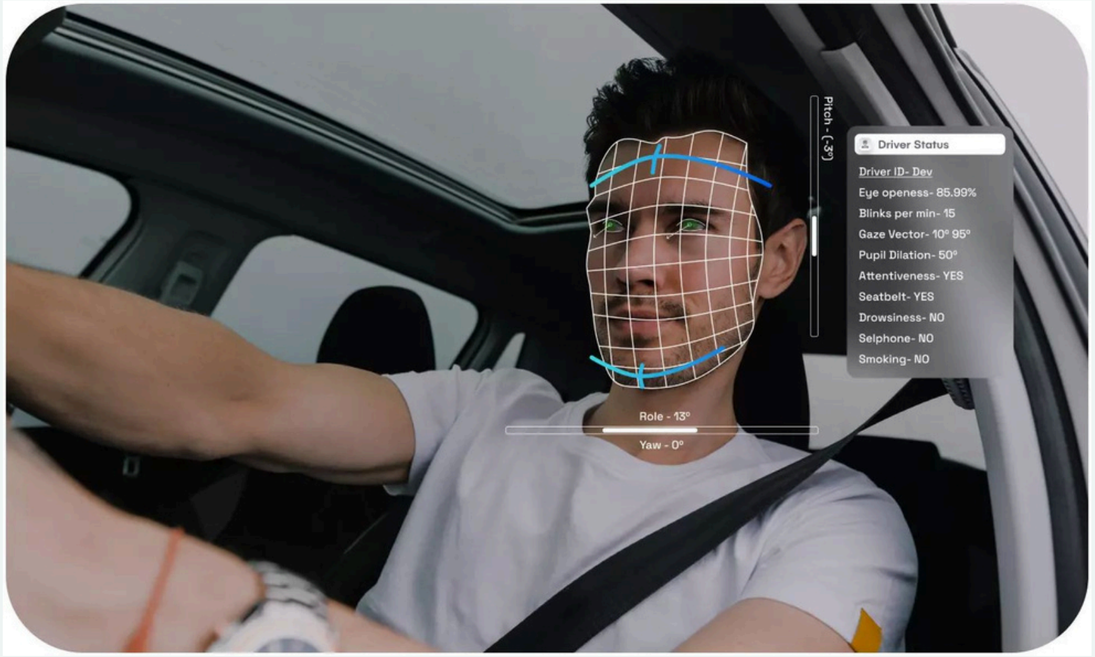
OBJECTIVES

- Detect drowsiness in real time
- Monitor eye movement and blinking
- Alert the driver immediately
- Build a low-cost and efficient system
- Improve road safety

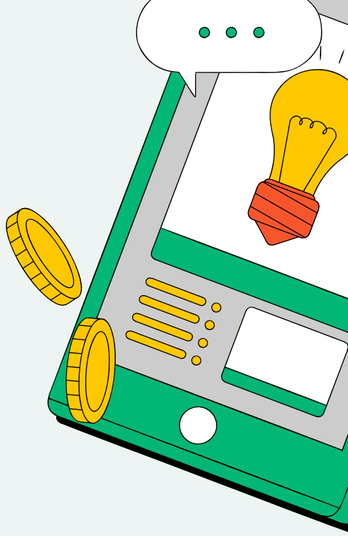


PROPOSED SYSTEM

“We built a real-time system that watches the driver's face through a webcam and automatically triggers an alert when drowsiness is detected — before an accident can happen.”



SYSTEM ARCHITECTURE



Camera captures live video

Frame extraction

Face detection

Feature extraction

Eye detection

Classification

Alert generation

TECHNOLOGIES USED

Python

OpenCV

Dlib / MediaPipe

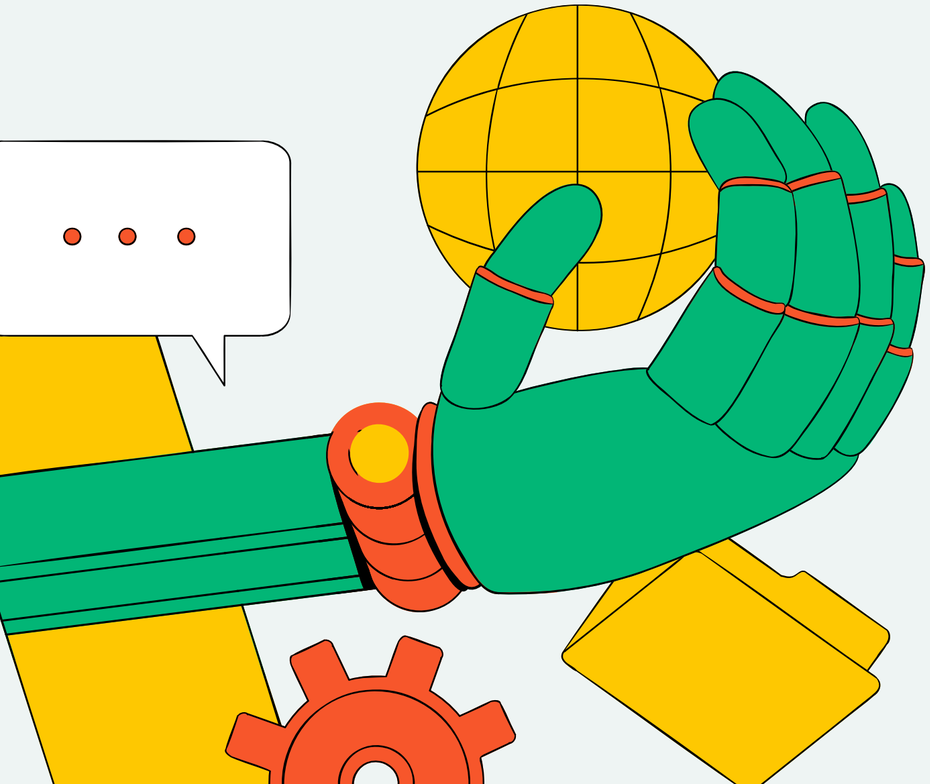
TensorFlow / Keras

Webcam / Alarm

ALGORITHMS USED

- Face Detection (Haar Cascade / CNN)
- Eye Detection (Facial Landmarks)
- CNN for classification
- Threshold-based alert system

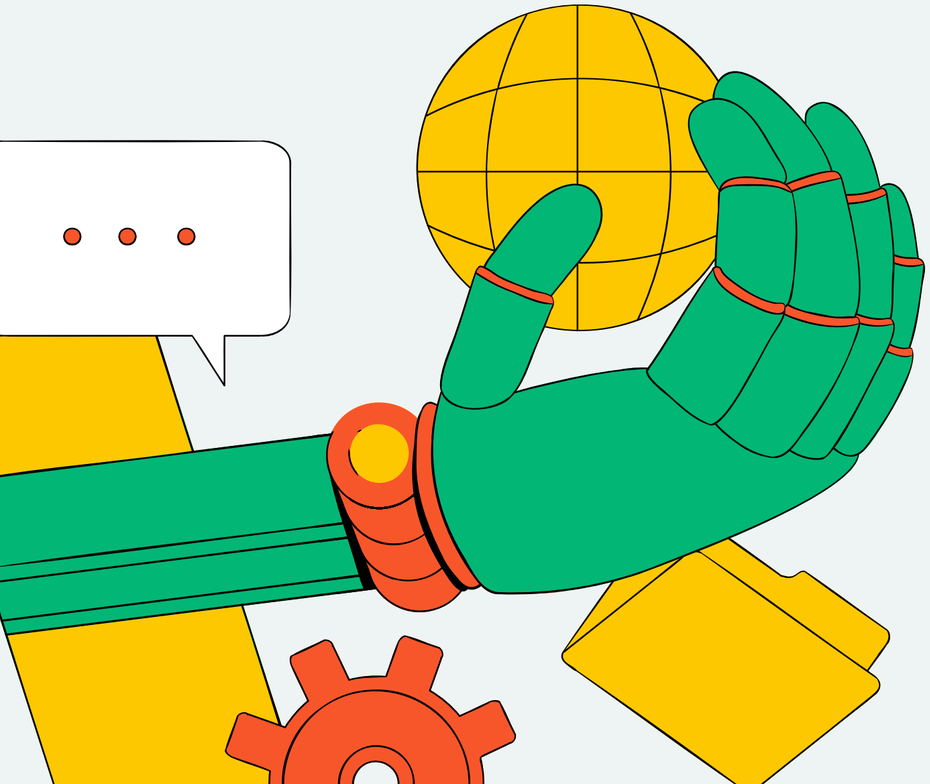
WORKING PRINCIPLE



The system captures video frames, detects the face and eyes, and calculates the Eye Aspect Ratio. If eyes remain closed for a certain duration, an alert is triggered.

“If your eyes close, our system opens them—virtually.”

DATASETS USED



- NTHU Driver Drowsiness Dataset
- Contains videos of drivers in different conditions such as:
  - Normal driving
  - Drowsy driving

  Includes variations in:
  - Lighting (day/night)
  - Head movement
  - Facial expressions

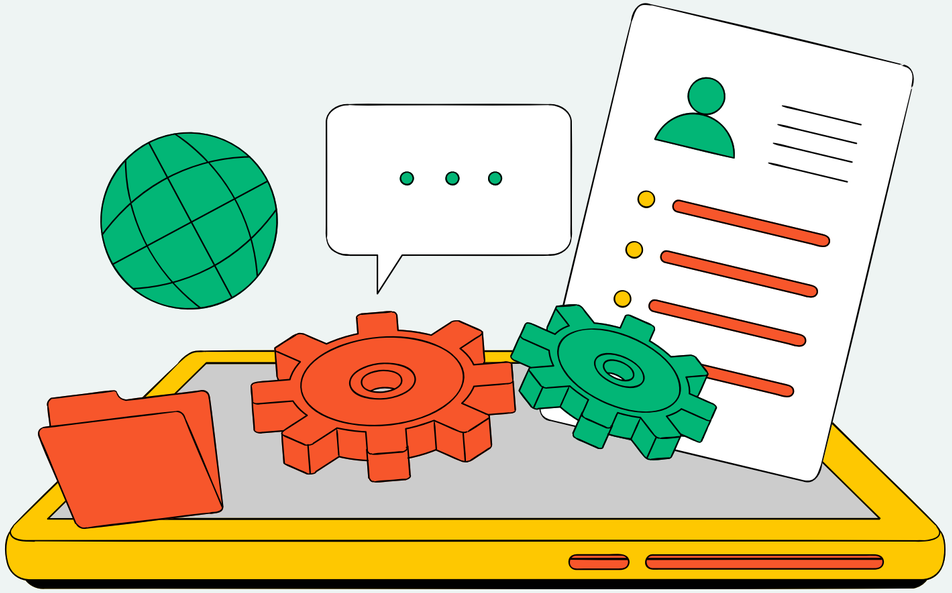
- Yawn Detection DatasetContains images of drivers with:
  - Open mouth (yawning)
  - Closed mouth (normal state)

RESULTS / OUTPUT

Real-time detection

Instant alert system

High accuracy



ADVANTAGES

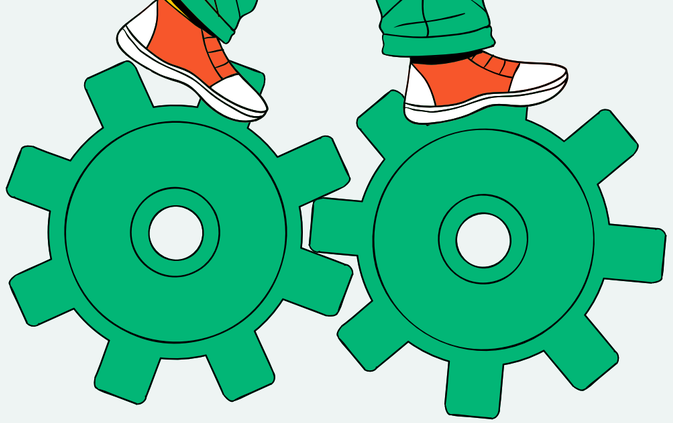
PREVENTS ACCIDENTS

REAL-TIME MONITORING

AFFORDABLE

EASY INTEGRATION

LIMITATIONS



1.

Sensitive to lighting conditions

2.

Glasses may affect accuracy

3.

Possible false alerts

4.

Requires camera setup

FUTURE ENHANCEMENTS

Infrared cameras for night

Yawning detection

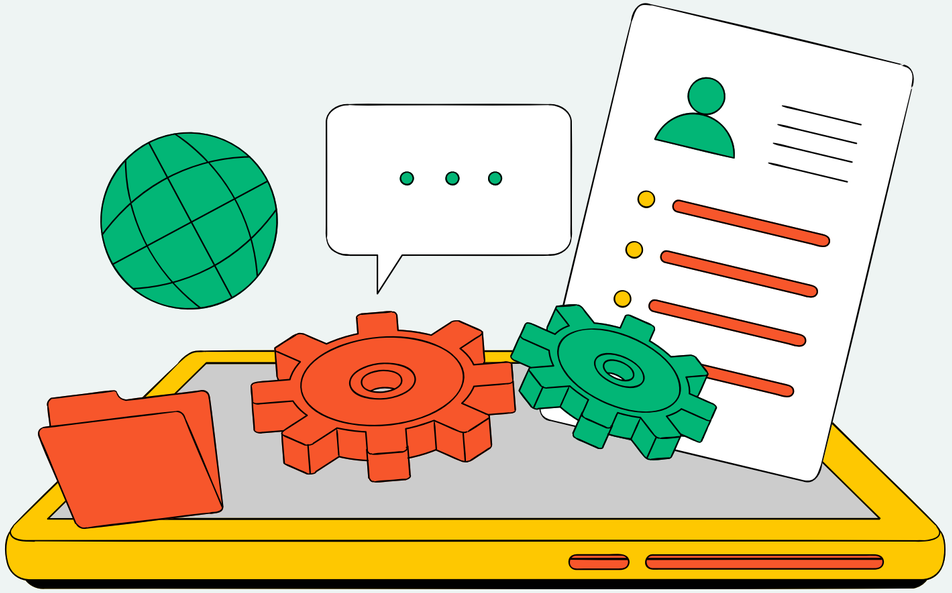
Mobile app alerts

APPLICATIONS

Public transport

Industrial vehicles

Aviation

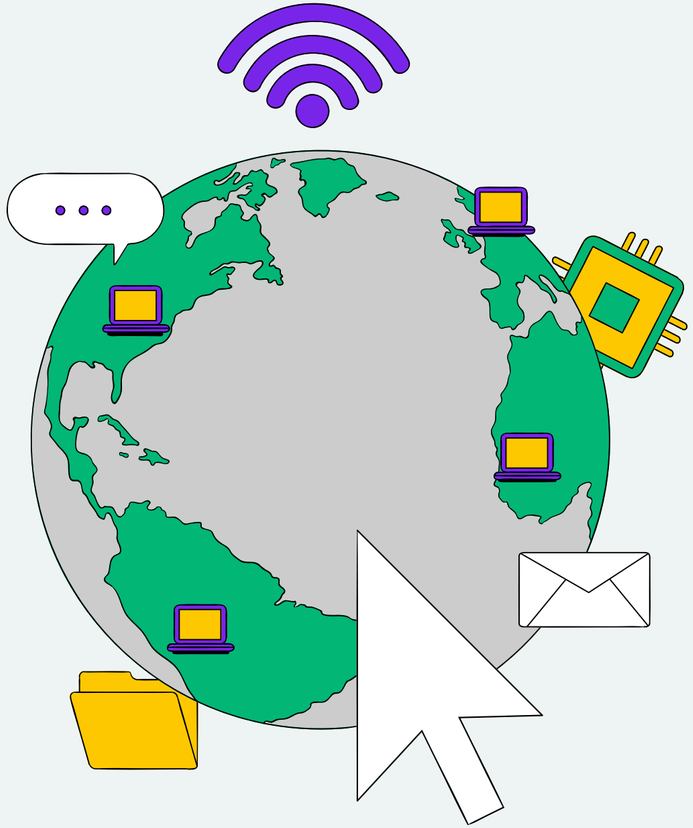


SOCIAL IMPACT

- Reduces road accidents caused by driver fatigue
- Provides real-time alerts to prevent dangerous situations
- Enhances safety for drivers and passengers

- USEFUL FOR LONG-DISTANCE AND COMMERCIAL DRIVER
- COST-EFFECTIVE AND WIDELY ACCESSIBLE SOLUTION

CONCLUSION



The AI-based driver drowsiness detection system is an effective solution to reduce road accidents caused by fatigue. By combining computer vision and machine learning techniques, the system can monitor the driver's condition in real time and provide timely alerts when signs of drowsiness are detected.

This project demonstrates how artificial intelligence can be applied to solve real-world problems and improve safety in everyday life.

Overall, this technology can play a vital role in making transportation safer and smarter, helping to protect lives and prevent accidents.

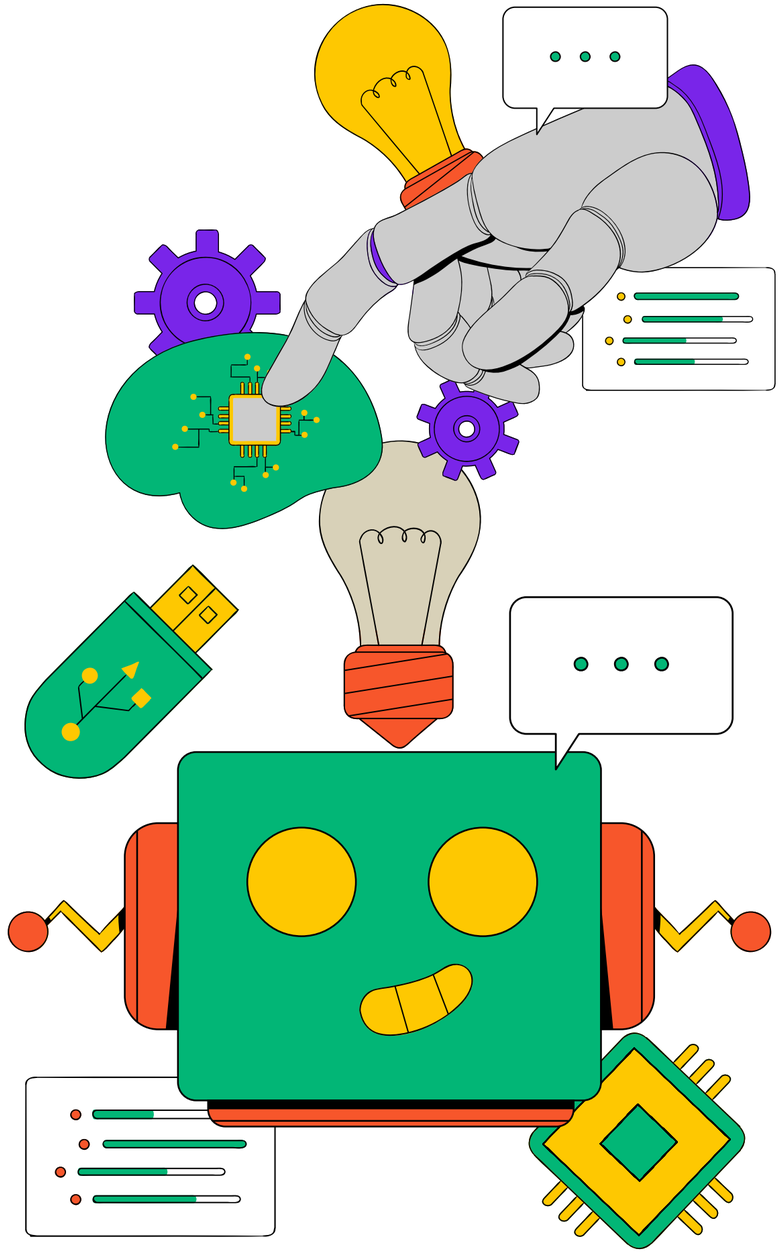
REFERENCES

Built on research, designed for reality.”

OpenCV Documentation	TensorFlow
IEEE Papers	Google Scholar

QUESTIONS AND ANSWERS

Your insights and questions are highly valuable to us, and we want to create an engaging and interactive session. Please feel free to send us your questions and concerns for clarifications.



THANK

YOU